

SUPPLEMENTARY MATERIALS

Thes supplementary materials sections are structured as follows: section. A details proposition related to the $\mathbf{g}$ update, section. B details proposition related the $\boldsymbol{\theta}$ update, and section. C details proposition related to identifiability requirement.

A. $\mathbf{g}$ Update

Proposition 1. Let $f_1(\mathbf{g}) = \|\mathbf{Y} - \mathbf{H}_1\mathbf{g} - \mathbf{B}_1\mathbf{d}\|_F^2$, $f_2(\mathbf{g}) = \lambda_1\|\mathbf{H}_2\mathbf{g} - \tilde{\boldsymbol{\eta}}\|_F^2$, $f_3(\mathbf{g}) = \lambda_2\|\mathbf{g}\|_1$, and $f_4(\mathbf{g}) = \lambda_3\sum_{q=1}^{(q_1+q_3)}\|\mathbf{g}_q\|_2$. The proximal operators of proximal operators of $\Delta\mathbf{f}_1, \dots, \Delta\mathbf{f}_4$ are given as follows: Let $\mathbf{z}$ and $\mathbf{a}$ have the same size as $\mathbf{g}$.

- 1) If $\mathbf{z} = \text{prox}_{\Delta\mathbf{f}_1}[\mathbf{a}]$, we have

$$\text{prox}_{\Delta\mathbf{f}_1}[\mathbf{a}] = (\mathbf{I} + \Delta\mathbf{H}_1^\top\mathbf{H}_1)^{-1}(\mathbf{a} + \Delta\mathbf{H}_1^\top(\mathbf{Y} - \mathbf{B}_1\mathbf{d})),$$
where $\mathbf{I}$ is the identity matrix of order p .
- 2) If $\mathbf{z} = \text{prox}_{\Delta\mathbf{f}_2}[\mathbf{a}]$, we have

$$\text{prox}_{\Delta\mathbf{f}_2}[\mathbf{a}] = (\mathbf{I} + \lambda_1\Delta\mathbf{H}_2^\top\mathbf{H}_2)^{-1}(\mathbf{a} + \lambda_1\Delta\mathbf{H}_2^\top\tilde{\boldsymbol{\eta}}),$$
where $\mathbf{I}$ is the identity matrix of order p .
- 3) If $\mathbf{z} = \text{prox}_{\Delta\mathbf{f}_3}[\mathbf{a}]$, we have

$$\text{prox}_{\Delta\mathbf{f}_3}[\mathbf{a}] = S_{\lambda_2\Delta}(\mathbf{a}) = \begin{cases} \mathbf{a} - \lambda_2\Delta, & \mathbf{a} > \lambda_2\Delta \\ \mathbf{0}, & -\lambda_2\Delta \leq \mathbf{a} \leq \lambda_2\Delta \\ \mathbf{a} + \lambda_2\Delta, & \mathbf{a} < -\lambda_2\Delta \end{cases}$$
- 4) If $\mathbf{z} = \text{prox}_{\Delta\mathbf{f}_4}[\mathbf{a}]$, we have

$$\text{prox}_{\Delta\mathbf{f}_4}[\mathbf{a}] = \left[1 - \frac{\lambda_3\Delta}{\|\mathbf{a}\|_2}\right]_+ \mathbf{a}.$$

Proof

The proximal operator of $\Delta\mathbf{f}_1$: The $f_1(\mathbf{g})$ can be written as a least square form as follows:

$$f_1(\mathbf{g}) = \|\mathbf{Y} - \mathbf{H}_1\mathbf{g} - \mathbf{B}_1\mathbf{d}\|_F^2 = \|(\mathbf{Y} - \mathbf{B}_1\mathbf{d}) - \mathbf{H}_1\mathbf{g}\|_F^2.$$

According to [46], the proximal operator for $q(x) = \|\mathbf{c} - \mathbf{P}\mathbf{x}\|_2^2$ can be written as follows:

$$\text{prox}_{\Delta q}[\mathbf{x}] = \underset{\mathbf{t}}{\text{argmin}} \left\{ \Delta q(\mathbf{t}) + \frac{1}{2}\|\mathbf{t} - \mathbf{x}\|_2^2 \right\} \\ = (\mathbf{I} + \Delta\mathbf{P}^\top\mathbf{P})^{-1}(\mathbf{x} + \Delta\mathbf{P}^\top\mathbf{c})$$

In our setting, $\mathbf{P} = \mathbf{H}_1$, $\mathbf{c} = (\mathbf{Y} - \mathbf{B}_1\mathbf{d})$, hence we have,

$$\text{prox}_{\Delta\mathbf{f}_1}[\mathbf{a}] = (\mathbf{I} + \Delta\mathbf{H}_1^\top\mathbf{H}_1)^{-1}(\mathbf{a} + \Delta\mathbf{H}_1^\top(\mathbf{Y} - \mathbf{B}_1\mathbf{d}))$$

The proximal operator of $\Delta\mathbf{f}_2$: The $f_2(\mathbf{g})$ can be written in a similar format as $f_1(\mathbf{g})$, hence, by setting $\mathbf{P} = \mathbf{H}_2$, $\mathbf{c} = \tilde{\boldsymbol{\eta}}$, we have

$$\text{prox}_{\Delta\mathbf{f}_2}[\mathbf{a}] = (\mathbf{I} + \lambda_1\Delta\mathbf{H}_2^\top\mathbf{H}_2)^{-1}(\mathbf{a} + \lambda_1\Delta\mathbf{H}_2^\top\tilde{\boldsymbol{\eta}})$$

The proximal operator of $\Delta\mathbf{f}_3$: This can be obtained from the proximal operator of l_1 norm, and it is given in [46] and can be applied in our setting as follows:

$$\text{prox}_{\Delta\mathbf{f}_3}[\mathbf{a}] = S_{\lambda_2\Delta}(\mathbf{a}) = \begin{cases} \mathbf{a} - \lambda_2\Delta, & \mathbf{a} > \lambda_2\Delta \\ \mathbf{0}, & -\lambda_2\Delta \leq \mathbf{a} \leq \lambda_2\Delta \\ \mathbf{a} + \lambda_2\Delta, & \mathbf{a} < -\lambda_2\Delta \end{cases}$$

The proximal operator of $\Delta\mathbf{f}_4$: This term is the sum of l_2 norms. By the separable property of the proximal operator, the result follows from the proximal operator of l_2 norms given in [46] as:

$$\text{prox}_{\Delta\mathbf{f}_4}[\mathbf{a}] = \left[1 - \frac{\lambda_3\Delta}{\|\mathbf{a}\|_2}\right]_+ \mathbf{a}$$

B. $\boldsymbol{\theta}$ Update

Proposition 2. Let $f_1(\mathbf{g}) = \|\mathbf{Y} - \mathbf{H}_1\mathbf{g} - \mathbf{B}_1\mathbf{d}\|_F^2$, $f_2(\mathbf{g}) = \lambda_1\|\mathbf{H}_2\mathbf{g} - \tilde{\boldsymbol{\eta}}\|_F^2$ and $f_3(\mathbf{g}) = \sum_k^{q_3} \mathbf{I}_{\omega_{Lk} \leq \theta_k \leq \omega_{Uk}}$. The proximal operators of proximal operators of $\Delta\mathbf{f}_1, \dots, \Delta\mathbf{f}_3$ are given as follows: Let $\mathbf{z}$ and $\mathbf{a}$ have the same size as $\boldsymbol{\theta}$.

- 1) If $\mathbf{z} = \text{prox}_{\Delta\mathbf{f}_1}[\mathbf{a}]$, we have

$$\text{prox}_{\Delta\mathbf{f}_1}[\mathbf{a}] = (\mathbf{I} + \Delta\mathbf{g}\mathbf{g}^\top)^{-1}(\mathbf{a} + \Delta\mathbf{g}(\mathbf{Y} - \mathbf{B}_1\mathbf{d})^\top),$$
where $\mathbf{I}$ is the identity matrix of order p .
- 2) If $\mathbf{z} = \text{prox}_{\Delta\mathbf{f}_2}[\mathbf{a}]$, we have

$$\text{prox}_{\Delta\mathbf{f}_2}[\mathbf{a}] = (\mathbf{I} + \lambda_1\Delta\mathbf{g}\mathbf{g}^\top)^{-1}(\mathbf{a} + \lambda_1\Delta\mathbf{g}\tilde{\boldsymbol{\eta}}^\top),$$
where $\mathbf{I}$ is the identity matrix of order p .
- 3) If $\mathbf{z} = \text{prox}_{\Delta\mathbf{f}_3}[\mathbf{a}]$, we have

$$z_k = \text{prox}_{\Delta\mathbf{f}_3}[a_k] = \max\{\omega_{Lk}, \min\{a_k, \omega_{Uk}\}\}.$$

Proof

The proximal operator of $\Delta\mathbf{f}_1$: The $f_1(\boldsymbol{\theta})$ can be written as a least square form and the proximal operator can be found similar to f_1 in proposition 1 as follows:

By setting, $\mathbf{P} = \mathbf{g}^\top$, $\mathbf{c} = (\mathbf{Y} - \mathbf{B}_1\mathbf{d})^\top$, $\mathbf{x} = \mathbf{H}_1^\top$, we have

$$\text{prox}_{\Delta\mathbf{f}_1}[\mathbf{a}] = (\mathbf{I} + \Delta\mathbf{g}\mathbf{g}^\top)^{-1}(\mathbf{a} + \Delta\mathbf{g}(\mathbf{Y} - \mathbf{B}_1\mathbf{d})^\top)$$

The proximal operator of $\Delta\mathbf{f}_2$: The $f_2(\boldsymbol{\theta})$ can be written in a similar format as $f_1(\mathbf{g})$, hence, by setting $\mathbf{P} = \mathbf{g}^\top$, $\mathbf{c} = \tilde{\boldsymbol{\eta}}^\top$, $\mathbf{x} = \mathbf{H}_2^\top$, we have

$$\text{prox}_{\Delta\mathbf{f}_2}[\mathbf{a}] = (\mathbf{I} + \lambda_1\Delta\mathbf{g}\mathbf{g}^\top)^{-1}(\mathbf{a} + \lambda_1\Delta\mathbf{g}\tilde{\boldsymbol{\eta}}^\top)$$

The proximal operator of $\Delta\mathbf{f}_3$: By the separable property of the proximal operator, the result follows from the proximal operator of box constraints given in [46] as follows:

$$z_k = \text{prox}_{\Delta\mathbf{f}_3}[a_k] = \max\{\omega_{Lk}, \min\{a_k, \omega_{Uk}\}\}$$

Another aspect of our algorithm that needs to be discussed is identifiability issues that may arise in evaluating terms on the form $\mathbf{H}_1\mathbf{g}$ appearing in (8) for instance. This problem can be cast as a dictionary learning problem since we are in some sense trying to learn a basis matrix, $\mathbf{H}_1$, and a sparse coefficients vector, $\mathbf{g}$. The success of the dictionary learning method does not require that the patterns in $\mathbf{H}_1$ to be orthogonal like PCA, and the reconstructed signal is often evaluated accurately under mild conditions. However, this does not guarantee that dictionary learning can uniquely recover sparse data from samples. Meaning, the uniqueness of the learned basis and sparse coefficients vector is not always guaranteed which may create an identifiability issue. The detailed statistical assumptions that guarantee the identifiability is stated in proposition 3 and further discussed in Appendix C.

C. Identifiability Requirement

Proposition 3: Let assume $\mathbf{g}$ is a vector with at most $k \ll m$ nonzero entries, called k -sparse. Assume the basis $\mathbf{H}$ is Spark on the vectors $\mathbf{g}$ such that:

$$\mathbf{H}\mathbf{g}_1 = \mathbf{H}\mathbf{g}_2 \text{ for } \mathbf{g}_1, \mathbf{g}_2 \in \mathbb{R}^m \Rightarrow \mathbf{g}_1 = \mathbf{g}_2$$

Assuming $\mathbf{H}$ is given, and our signal is constructed such that:

$$\tilde{\mathbf{Y}} = \mathbf{H}\mathbf{g}$$

where $\tilde{\mathbf{Y}} = \mathbf{Y} - \mathbf{B}_1\mathbf{d}$

a lower bound can be set on the number of samples, n , of $\tilde{\mathbf{Y}}$ that guarantees unique recovery of sparse vector $\mathbf{g}$ from a signal $\tilde{\mathbf{Y}}$ with a given basis matrix $\mathbf{A}$ such that:

$$n \geq Ck \log\left(\frac{m}{k}\right)$$

where C is a constant independent of m, n , and k with more detailed discussions and proofs included in [1].

In the case $\mathbf{H}$ is unknown, assuming the above holds, we assume we can subsample N similar samples of $\tilde{\mathbf{Y}}$ from our system such that we have $\tilde{\mathbf{Y}} = \{\tilde{\mathbf{Y}}_1, \dots, \tilde{\mathbf{Y}}_N\}$. Then, can guarantee the uniqueness of both $\mathbf{g}$ and $\mathbf{H}$ if we can guarantee a lower bound on the number of required subsamples N to guarantee uniqueness of $\mathbf{A}$ with probability $(1 - \beta)$ as follows:

$$N = \frac{k+1}{\beta} \binom{m}{k}$$

Proof:

Consider problem (7) and (8) in our formulation, since λ_1 control how much we emphasize on the physics-based model and can be considered as a regularization term, without loss of generality, we consider the case where $\lambda_1 \approx 0$, and reformulated the two problems as follows:

$$\min_{\mathbf{g}, \theta} \left\| \mathbf{Y} - \mathbf{H}_1 \mathbf{g} - \mathbf{B}_1 \mathbf{d} \right\|_F^2 + \lambda_2 \|\mathbf{g}\|_1 + \lambda_3 \sum_{q=1}^{(q_1+q_3)} \|\mathbf{g}_q\|_2$$

$$\Rightarrow \min_{\mathbf{g}, \theta} \left\| \tilde{\mathbf{Y}} - \mathbf{H}_1 \mathbf{g} \right\|_F^2 + \lambda_2 \|\mathbf{g}\|_1 + \lambda_3 \sum_{q=1}^{(q_1+q_3)} \|\mathbf{g}_q\|_2,$$

where $\tilde{\mathbf{Y}} = \mathbf{Y} - \mathbf{B}_1 \mathbf{d}$.

Our main concern here is under what assumptions when we evaluate our signal $\tilde{\mathbf{Y}}$, we can assume guarantees about the uniqueness of the learned basis $\mathbf{H}_1$ and sparse coefficients vector $\mathbf{g}$. Assume that observed data $\tilde{\mathbf{Y}}$ were generated as:

$$\tilde{\mathbf{Y}} = \mathbf{A} \mathbf{a} \quad (14)$$

using a fixed $n \times m$ basis matrix $\mathbf{A}$ and sparse m dimensional latent representations $\mathbf{a}$. First, we address the accuracy of the recovered $\mathbf{a}$, which mainly concerns the $\mathbf{g}$ update in Algorithm 1 of our problem. This can be done using theory from compressed sensing (CS) [17], where they have investigated conditions under which the sparse latent representations, $\mathbf{a}$, can be recovered from data [1]. Given an m -dimensional vector $\mathbf{a}$ with at most $k \ll m$ nonzero entries. Such vectors $\mathbf{a}$ are called k -sparse. And given a known basis matrix $\mathbf{A}$, a typical condition is:

$$\mathbf{A} \mathbf{a}_1 = \mathbf{A} \mathbf{a}_2 \text{ for } k \text{ sparse } \mathbf{a}_1, \mathbf{a}_2 \in \mathbb{R}^m \Rightarrow \mathbf{a}_1 = \mathbf{a}_2 \quad (15)$$

known as the *spark condition*. Another condition is related to the number of samples, n , of $\tilde{\mathbf{Y}}$ needs to satisfy:

$$n \geq Ck \log\left(\frac{m}{k}\right), \quad (16)$$

where C is a constant independent of m, n , and k with more detailed discussions and proofs included in [1].

Conditions (17) and (18) describe when it is possible to recover a sparse vector $\mathbf{a}$ from a signal $\tilde{\mathbf{Y}}$ with a given or known basis matrix $\mathbf{A}$.

Second, when the $\mathbf{A}$ is unknown, it is more difficult to guarantee the uniqueness of both $\mathbf{a}$ and $\mathbf{A}$ and requires more

conditions. To show this, if $\mathbf{P}$ is a permutation matrix and $\mathbf{D}$ is an invertible diagonal matrix, then

$$\mathbf{A} \mathbf{a} = (\mathbf{A} \mathbf{D}^{-1} \mathbf{P}^T) (\mathbf{P} \mathbf{D} \mathbf{a}), \quad (17)$$

where $\mathbf{P}$ is a permutation matrix with binary entries and precisely one 1 in each row and column such that $\mathbf{P} \mathbf{P}^T = \mathbf{P}^T \mathbf{P} = \mathbf{I}$. Hence, if $\mathbf{A}$ is unknown, we cannot discriminate which of $\mathbf{a}$ or $\mathbf{P} \mathbf{D} \mathbf{a}$ (respectively, $\mathbf{A}$ or $\mathbf{A} \mathbf{D}^{-1} \mathbf{P}^T$) was the original sparse vector or basis matrix, respectively. Suppose that the true signal $\tilde{\mathbf{Y}}$ is created according to (14) and we are trying to evaluate it as follows:

$$\tilde{\mathbf{Y}} = \mathbf{B} \mathbf{b} \quad (18)$$

So, we can guarantee the uniqueness of both $\mathbf{a}$ and $\mathbf{A}$ if we can guarantee that $\mathbf{A} \mathbf{a} = \mathbf{B} \mathbf{b}$ under scenario such that in (17). Suppose in our setting that $\tilde{\mathbf{Y}}$ is sampled from the system and we can subsample N similar samples such that we have $\tilde{\mathbf{Y}} = \{\tilde{\mathbf{Y}}_1, \dots, \tilde{\mathbf{Y}}_N\}$. Now assuming each subsample maybe associated with different active patterns in the basis matrix $\mathbf{A}$ hence may generate different vectors $\mathbf{a}$ such that $\tilde{\mathbf{a}} = \{\mathbf{a}_1, \dots, \mathbf{a}_N\}$. Assuming (15) and (16) hold, a lower bound on the number of required subsamples N can be found in [28] to guarantee uniqueness of $\mathbf{A}$ with probability $(1 - \beta)$ as follows:

$$N = \frac{k+1}{\beta} \binom{m}{k} \quad (19)$$

This indicates that sparse matrix factorization of basis matrix $\mathbf{A}$ is unique for a large enough number of subsamples.